Xiaobo Feng

23300180054@m.fudan.edu.cn +86 198 8492 2249 Shanghai, China

Education

Fudan University Sep 2023 – Jun 2027 (Expected)
B.S. in Mathematics and Applied Mathematics

- Cumulative GPA: 3.92/4.0; Rank: 4/226
- Selected Coursework: Introduction to Algebraic Number Theory (A), Advanced Algebraic Number Theory (A), Introduction to Modular Forms (A)

Research Interests

Arithmetic Geometry; p-adic Geometry; Algebraic Number Theory.

Honors & Awards

- **National Scholarship** (2025, 2026)
- **Bronze Medal in Analysis and PDE**, S.-T. Yau College Student Mathematics Contest (2026)
- **Honorable Mention in Algebra, Number Theory and Combinatorics**, S.-T. Yau College Student Mathematics Contest (2026)
- **Team Honorable Mention**, S.-T. Yau College Student Mathematics Contest (2026)
- Fudan University First-Class Scholarship (Jiahui Education Scholarship) (2024)
- Fudan University Foundational Discipline Scholarship (2024, 2025)
- First Prize, Shanghai Division, Chinese Mathematics Competition (2024, 2025)

Academic Activities & Reading

Independent Study in p-adic Geometry Sep 2025 – Present
Advisor: Prof. Haining Wang, Shanghai Center for Mathematical Sciences

- **Phase I: Formal and Rigid Geometry.** Studied the foundations of formal and rigid analytic geometry, with particular emphasis on Drinfeld's p-adic upper half-plane.
  Main reference: S. Bosch, *Lectures on Formal and Rigid Geometry.*
- **Phase II (Current): Drinfeld's Theorem and Perfectoid Geometry.** Studying the moduli interpretation of Drinfeld's p-adic upper half-plane, its connections with p-divisible groups and p-adic uniformization, as well as the theory of perfectoid spaces.
  Main references:
  - J.-F. Boutot and H. Carayol, *Uniformisation p-adique des courbes de Shimura : les théorèmes de Čerednik et de Drinfeld.* (Parts I and II completed.)
  - P. Scholze and J. Weinstein, *Berkeley Lectures on p-adic Geometry* (in progress).

Algebraic Geometry Seminar Mar 2025 – Present
School of Mathematical Sciences, Fudan University (Top-Notch Undergraduate Training Program)

- Systematically studying Ravi Vakil's *The Rising Sea: Foundations of Algebraic Geometry*; completed Chapters 1–18 (out of 29).
- Participate in weekly presentations and discussions, preparing lecture notes and blackboard presentations on selected chapters.

Summer School in Algebra and Number Theory (Seminar) Jul 2026 – Aug 2026
Academy of Mathematics and Systems Science, CAS & School of Mathematical Sciences, Peking University

- Participated in a six-week reading seminar on Schneider–Teitelbaum's work concerning representations associated with the cohomology of sheaves of differential forms on Drinfeld's p-adic symmetric spaces, under the guidance of Dr. Benchao Su (Ph.D., Peking University).

- Selected references:
  - P. Schneider and J. Teitelbaum, *An Integral Transform for p-adic Symmetric Spaces.*
  - P. Schneider and J. Teitelbaum, *p-adic Boundary Values.*

Summer School in Algebra and Number Theory (Advanced Track) Jul 2025 – Aug 2025
Academy of Mathematics and Systems Science, CAS & School of Mathematical Sciences, Peking University

- Participated in a six-week advanced program covering algebraic number theory, representation theory, and algebraic geometry.
- Awarded the "Excellent Participant" certificate.

Real Analysis Reading Seminar Jan 2024 – Jul 2024
School of Mathematical Sciences, Fudan University (Top-Notch Undergraduate Training Program)

- Completed a collective reading of the first six chapters of Gerald Folland's *Real Analysis: Modern Techniques and Their Applications.*
- Topics covered: measure theory, L^p spaces, abstract integration, and the Radon–Nikodym theorem.

Academic Service

Founder & Lead Instructor, "Shu Hai Qi Hang" Seminar Series Mar 2024 – Present
Student Learning and Development Center, School of Mathematical Sciences, Fudan University

- **Introduction to p-adic Numbers** (Oct 2024, Oct 2025):
  Organized two short seminar series (five sessions in total) and authored accompanying LaTeX lecture notes.
  Topics included constructions of p-adic fields via completion and inverse limits, valuations, Hensel's lemma, basic extensions of local fields, and elementary p-adic analysis. Mentored student presentations.
- **Classification of Quadratic Forms over $\mathbb{Z}$** (Mar 2025):
  Organized a three-session seminar based on Chapter V of Serre's *A Course in Arithmetic.* Translated and presented the material and guided students in preparing presentations.

Mathematics Tutor, "Shu Yuan Da Shen" Program Sep 2023 – Present
Student Learning and Development Center, School of Mathematical Sciences, Fudan University

- Provided weekly two-hour tutoring sessions in foundational mathematics courses, including Mathematical Analysis and Advanced Algebra.